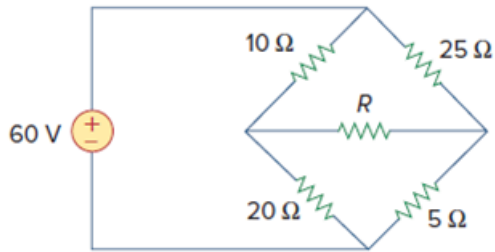


Problem

Determine the maximum power that can be delivered to the variable resistor R in the circuit of Fig. 4.139.

Figure. 4.139:



Step-by-step solution

Step 1 of 7

Refer to Figure 4.139 in the textbook.

To find the maximum power delivered to the variable resistor R , first determine the Thevenin equivalent circuit across the variable resistor.

The Thevenin equivalent resistance is the resistance seen by the open terminals when replacing all sources with their internal resistances. Draw the modified circuit diagram.

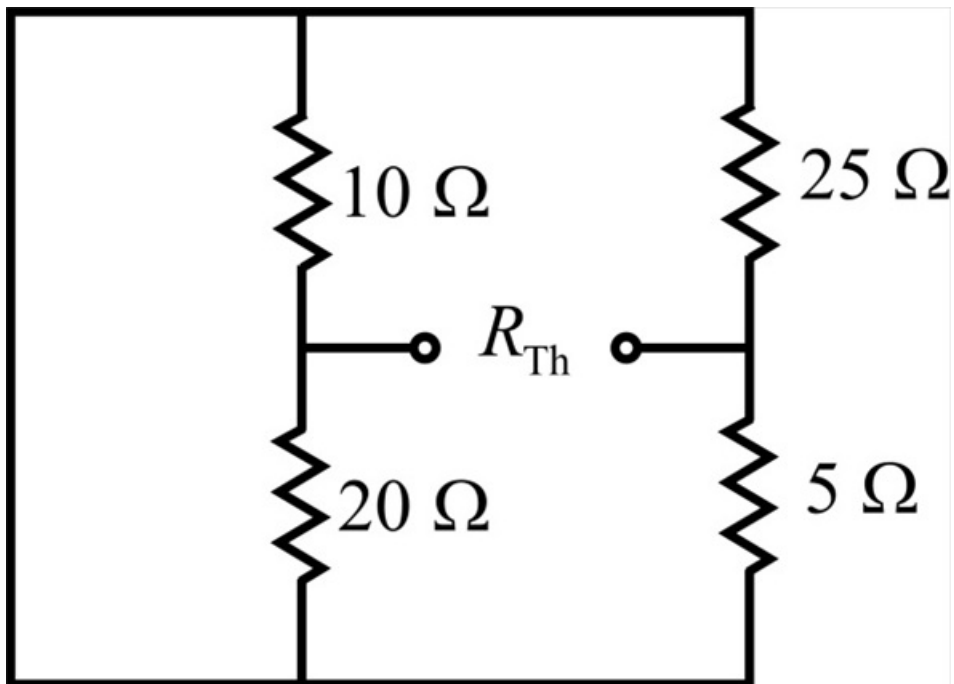


Figure 1

Step 2 of 7

In Figure 1, the resistors $10\ \Omega$ and $20\ \Omega$ are connected in parallel; similarly the resistors $25\ \Omega$ and $5\ \Omega$ are connected in parallel. These parallel combinations are in turn connected in series. Calculate its equivalent resistance.

$$\begin{aligned} R_{Th} &= (10 \parallel 20) + (25 \parallel 5) \\ &= \left(\frac{10 \times 20}{10 + 20} \right) + \left(\frac{25 \times 5}{25 + 5} \right) \\ &= \frac{20}{3} + \frac{25}{6} \\ &= 10.833\ \Omega \end{aligned}$$

Therefore, the Thevenin equivalent resistance is $R_{Th} = 10.833\ \Omega$.

Step 3 of 7

The Thevenin voltage is the voltage across open terminals. Draw the modified circuit diagram to find V_{Th} .

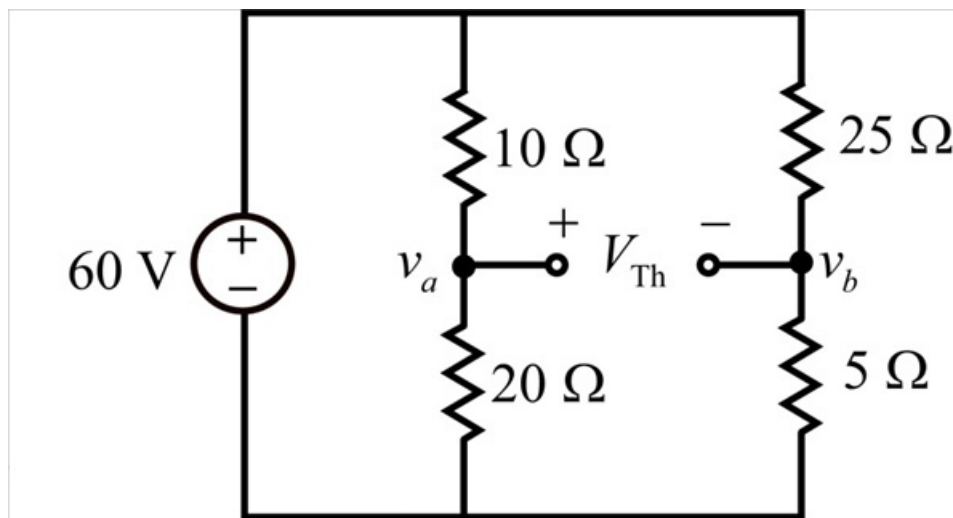


Figure 2

Step 4 of 7

From Figure 2, the expression for Thevenin voltage is,

$$V_{Th} = v_a - v_b$$

Find the value of node voltage v_a using voltage division principle.

$$\begin{aligned} v_a &= \left(\frac{20}{20 + 10} \right) (60) \\ &= \left(\frac{2}{3} \right) (60) \\ &= 40\text{ V} \end{aligned}$$

Find the value of node voltage v_b using voltage division principle.

$$\begin{aligned} v_b &= \left(\frac{5}{25 + 5} \right) (60) \\ &= \left(\frac{1}{6} \right) (60) \\ &= 10\text{ V} \end{aligned}$$

Step 5 of 7

Determine the Thevenin voltage V_{Th}

$$\begin{aligned} V_{Th} &= v_a - v_b \\ &= 40 - 10 \\ &= 30 \text{ V} \end{aligned}$$

Therefore, the Thevenin equivalent voltage is $V_{Th} = 30 \text{ V}$.

Step 6 of 7

Draw the Thevenin equivalent circuit across the variable resistor R .

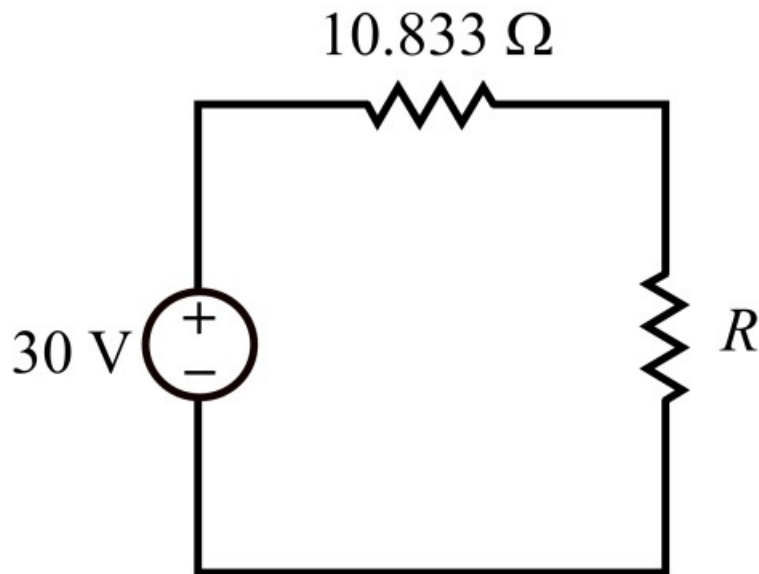


Figure 3

Step 7 of 7

The maximum power will be delivered to the variable resistor R , when it is equal to the Thevenin resistance. Therefore,

$$R = R_{Th}$$

Find the maximum power delivered to the variable resistor R .

$$p_{\max} = \frac{V_{Th}^2}{4R_{Th}}$$

Substitute 30 V for V_{Th} and 10.833 Ω for R_{Th} .

$$\begin{aligned} p_{\max} &= \frac{(30)^2}{4(10.833)} \\ &= 20.77 \text{ W} \end{aligned}$$

Therefore, the maximum power that can be delivered to the variable resistor in the circuit is

$$\boxed{20.77 \text{ W}}$$